

PROJECT 2

1. Why did you choose the classifier that you did?

I chose the Random Forest classifier because it works well for small datasets. It is good for data with features like mean, variance, min and max. It can handle complex patterns and does not need much preprocessing. It works well even if the data has noise. It also reduces overfitting by using multiple decision trees. As this project uses simple features from physiological signals, Random forest is a good and reliable choice for classification.

2. Which data type had the highest accuracy? Was it a data type that is commonly associated with pain? Describe why it is commonly associated with pain.

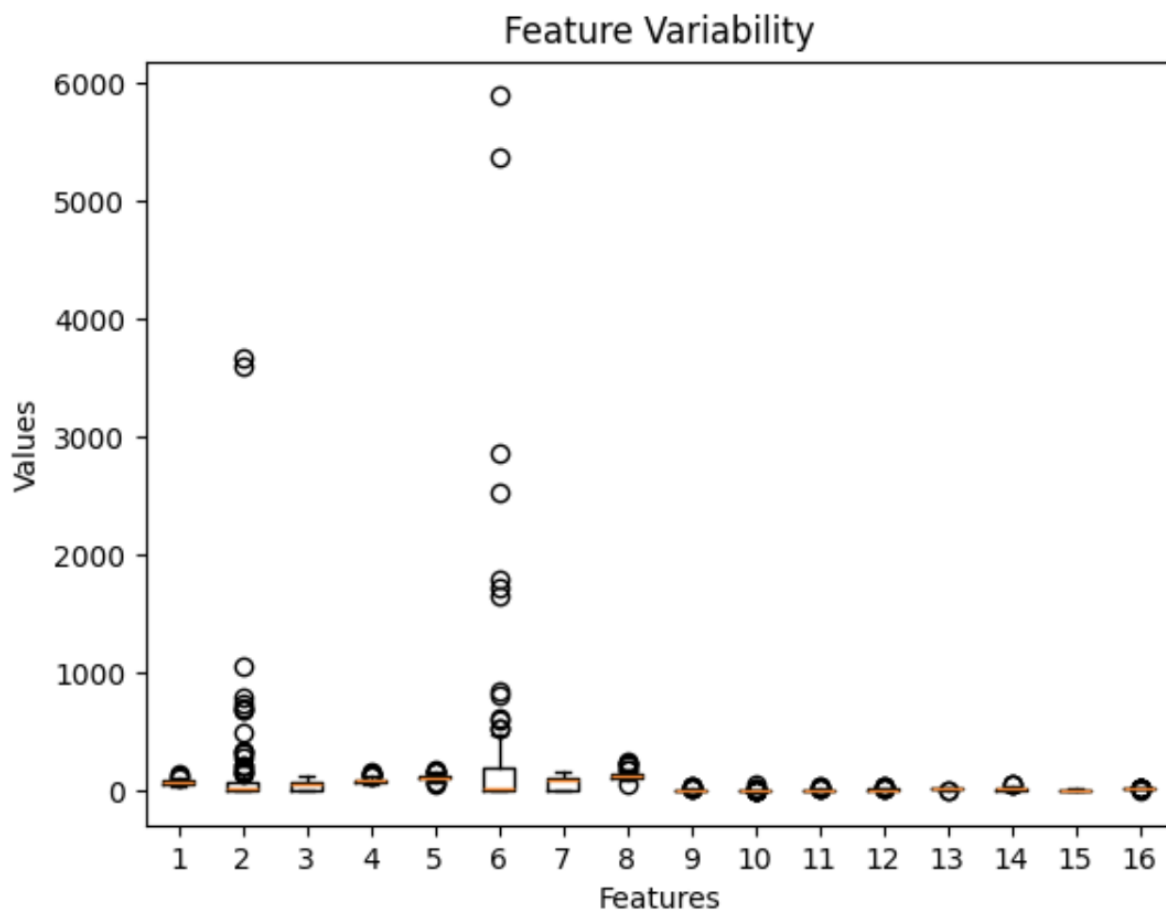
The data type with the highest accuracy is the fusion of all signals. It has an accuracy of 0.733, precision of 0.731, and recall of 0.75. This means the model works best when all signals are used together. This is expected because using multiple physiological signals gives more information about how the body responds to pain. Pain activates the autonomic nervous system. This affects signals like blood pressure, skin activity (EDA), and respiration. When all these signals are used together, the model can better find patterns related to pain. The confusion matrix for this data type is $\begin{bmatrix} 4.3 & 1.7 \\ 1.5 & 4.5 \end{bmatrix}$. This shows that most pain and no-pain cases are classified correctly. There are fewer errors.

3. Fusing data is a common approach in machine learning. How did your fusion features perform? If it had the highest accuracy why did this happen? If it was not the highest accuracy, why do you think this is the case ?

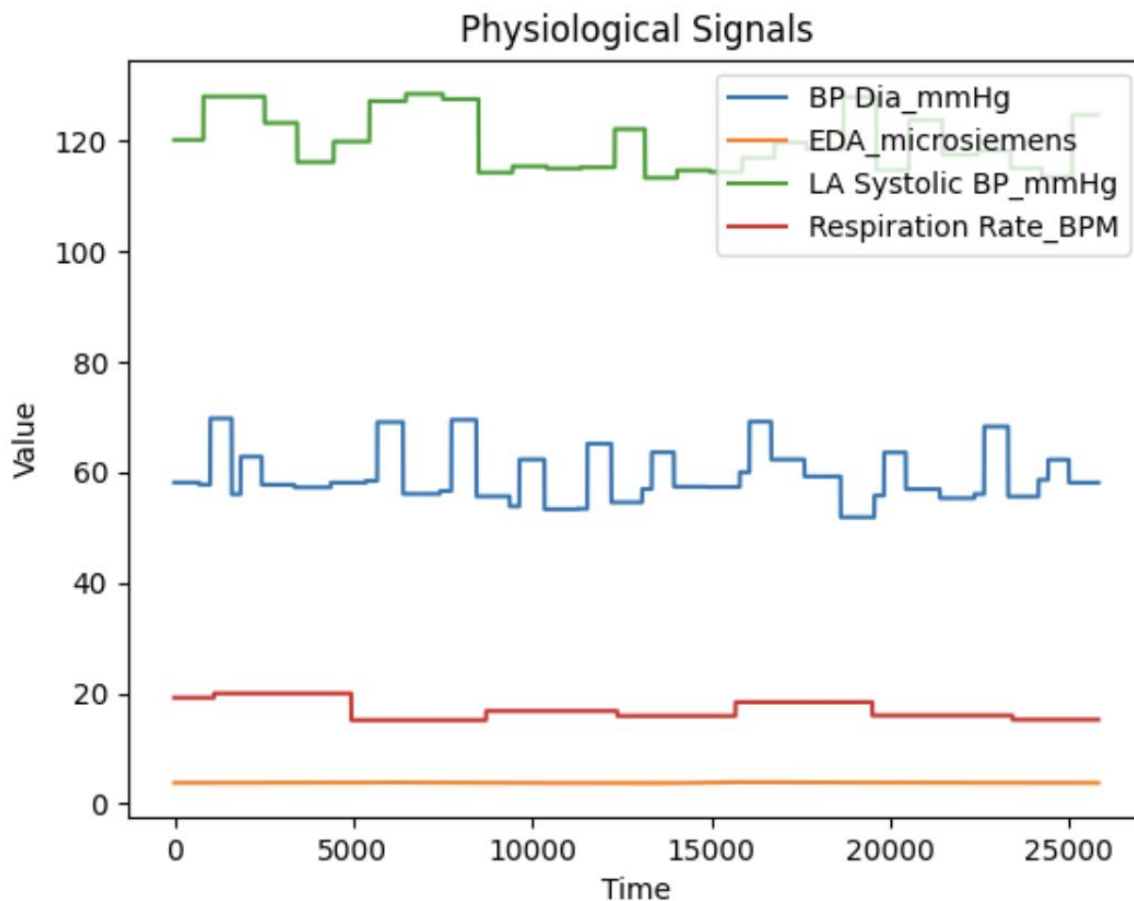
The fusion features “all” performed the best among all data types. It has the highest accuracy compared to individual signals. This shows that combining all signals improves the model performance. This happens because each physiological signal gives different information about the body. When these signals are combined, the model gets more complete information. This helps the model to better understand patterns related to pain. Fusion works well in machine learning because it reduces the limitation of using only one type of data. It helps to capture more features and improves prediction. In this project, combining blood pressure, EDA, and respiration helped to improve accuracy and recall.

4. Is there a lot of variability in the features that you created? Why do you think this is? To answer this, create a box plot that contains all the features. In other words, the plot will have 1 box for each feature type which will include lines coming from them that shows the variability of each feature.

Yes, there is variability in the features. This can be seen in the box plot, where the values are spread out and have different ranges. This happens because physiological signals change from person to person. They also depend on the level of pain. Different signals like blood pressure, EDA and respiration respond differently to pain. Because of this, the features like mean, variance, min and max also show variability.



5. Which physiological signal can visually be seen to have the most variability? To answer this, take a random instance of the original physiological signals and plot them on 1 line graph. Include a key to show which signal is which (can use different colors for each). Is the signal that looks like it has the most variability one that is commonly associated with pain. Give details about why you think it is or is not.



From the line plot, blood pressure shows more variability than EDA. EDA is almost flat and shows very little change. Systolic blood pressure shows the most change over time. Diastolic blood pressure also changes, but less. Respiration changes too, but less than blood pressure. Blood pressure is related to pain because pain activates the autonomic nervous system. This can increase blood pressure. Because of this, blood pressure can show clear changes during pain. So in this plot, the signal with the most variability is one that is commonly associated with pain.

6. There is some evidence that some physiological signals are correlated with facial movement (e.g., expressions) during levels of intense emotion (e.g. pain in this case). Give your thoughts and critiques on this. For this question, back up your answer with at least 1 citation from a published paper.

There is some evidence that pain is shown through both physiological changes and facial expressions. Pain affects both the body and a person's behavior. When a person feels pain, it causes changes in physiological signals like heart rate, respiration, and skin activity. It also causes facial expressions like eye tightening, brow lowering, and mouth opening (Hadjistavropoulos et al., 2007). These signals are connected because both are controlled by the nervous system. Pain activates the autonomic nervous system. This changes physiological signals and also causes facial muscle movements. Studies show that facial expressions are important indicators of pain (Hadjistavropoulos et al., 2007). However, there are some limitations. Facial expressions can be controlled or hidden, so they may not always show true pain. Physiological signals can also vary from person to person. They can be affected by factors like stress or emotion. In conclusion, physiological signals and facial expressions are both useful for detecting pain. Using them together gives better and more accurate results.

References : Hadjistavropoulos, T., et al. (2007). An interdisciplinary expert consensus statement on assessment of pain in older persons. The Clinical Journal of Pain.